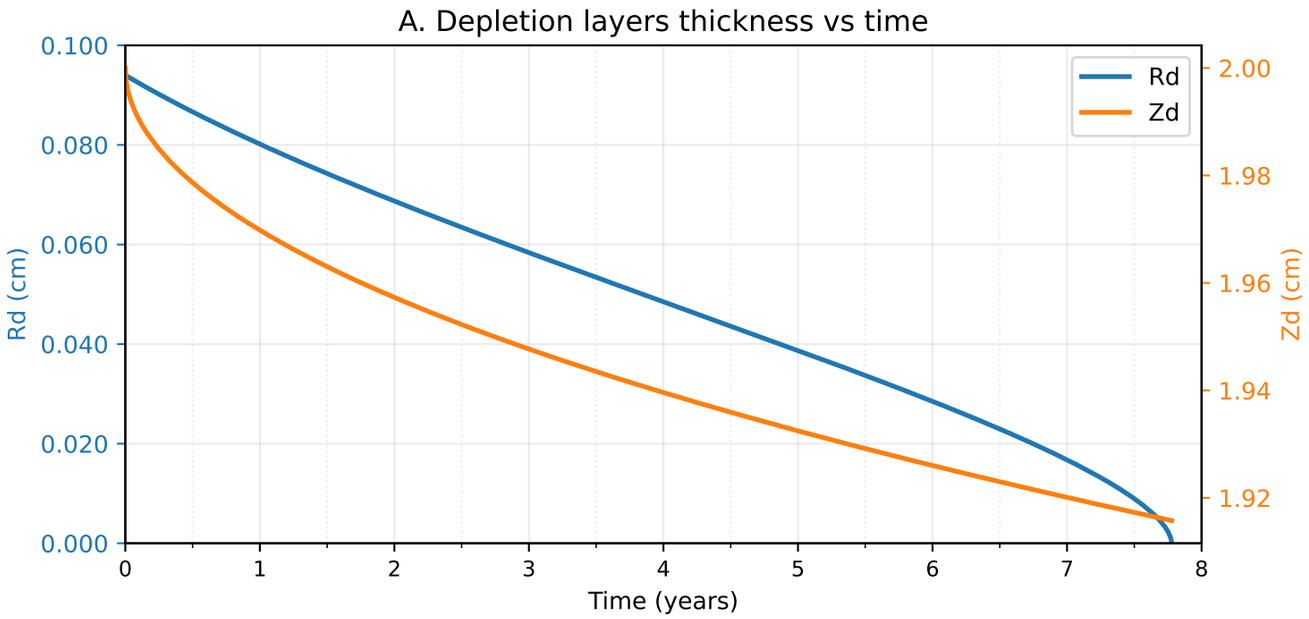
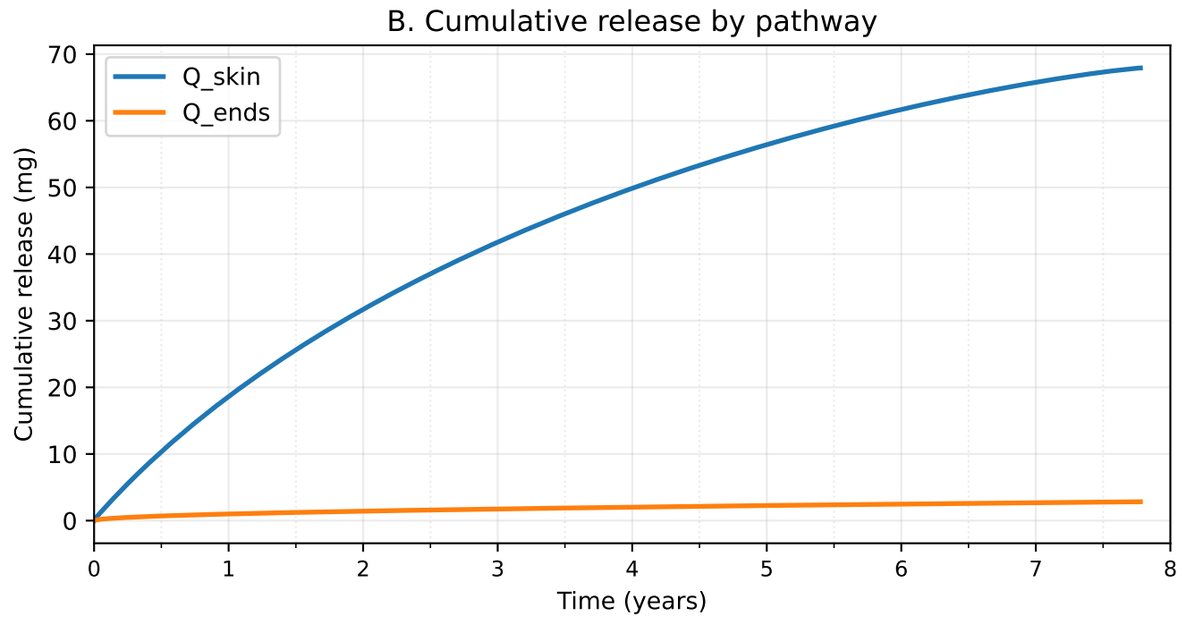
Nexplanon drug release model

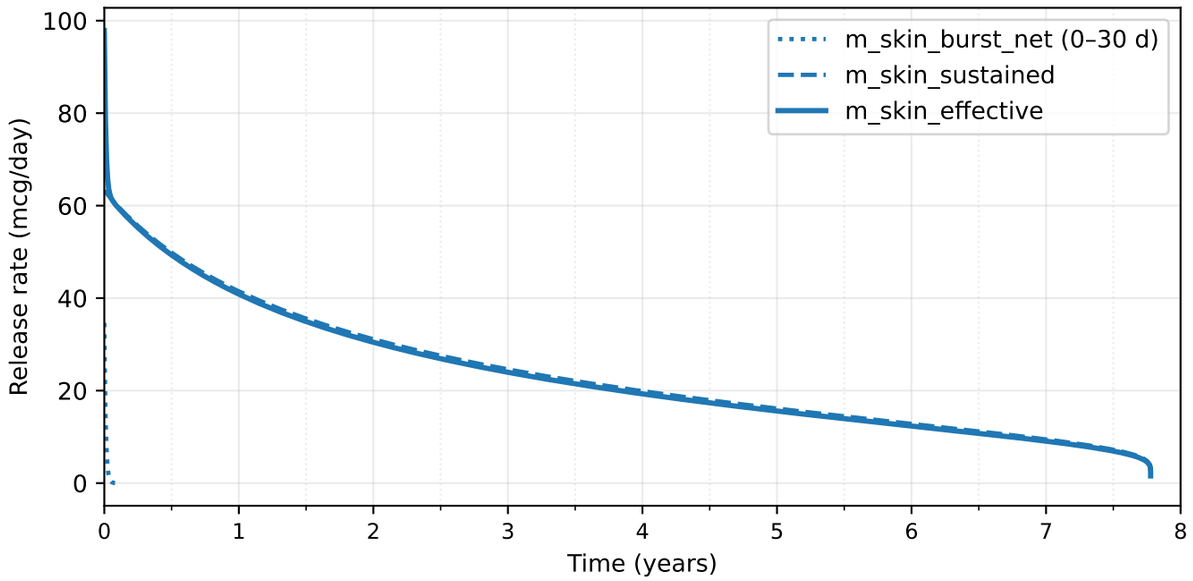
PSS model fitting report

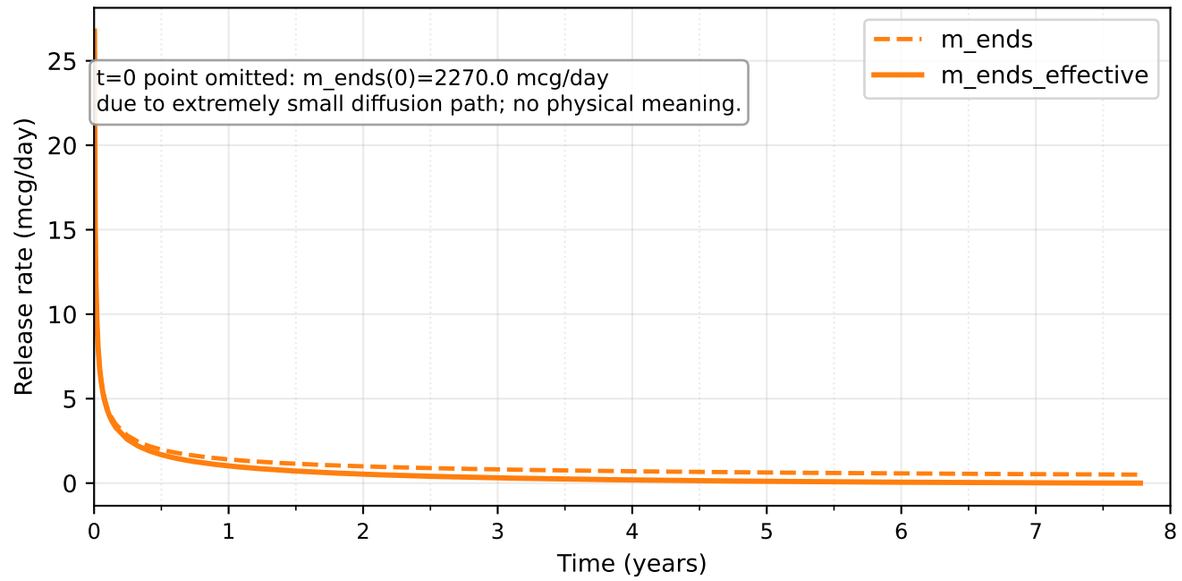
Organon release data - fitted DOS, PDL, ha_skin, and ha_ends

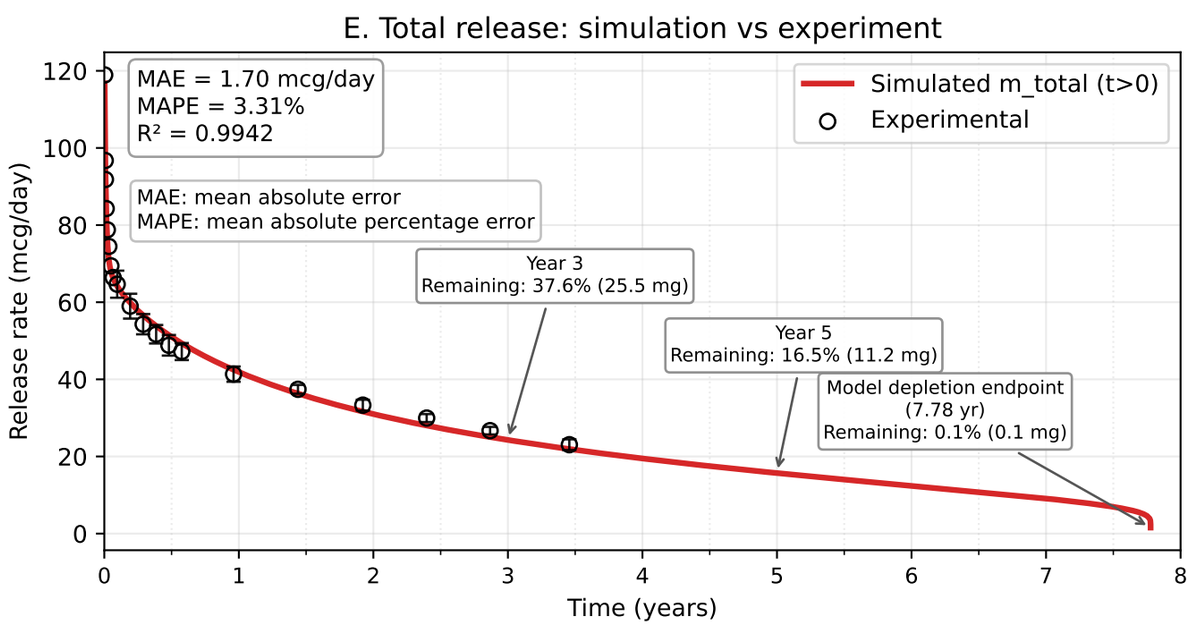
Figure 1. Paul model fitting diagnostics
DOS=1.535, PDL=9.043e-12 g/(cm·s), ha_skin=0.07808 cm, ha_ends=0.000100 cm











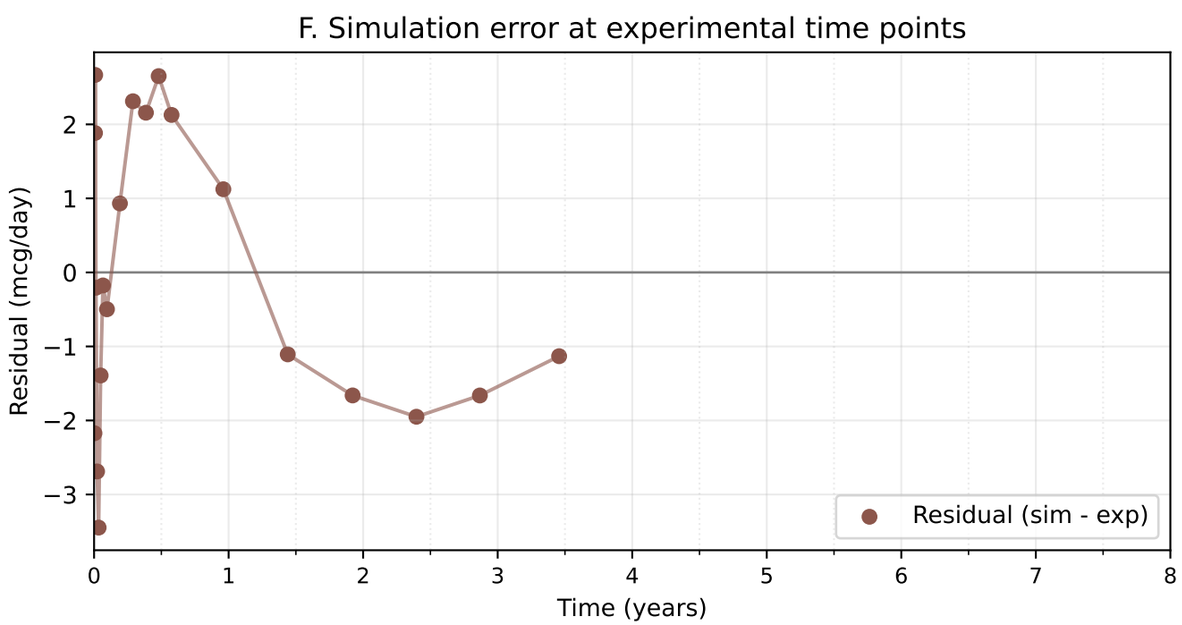
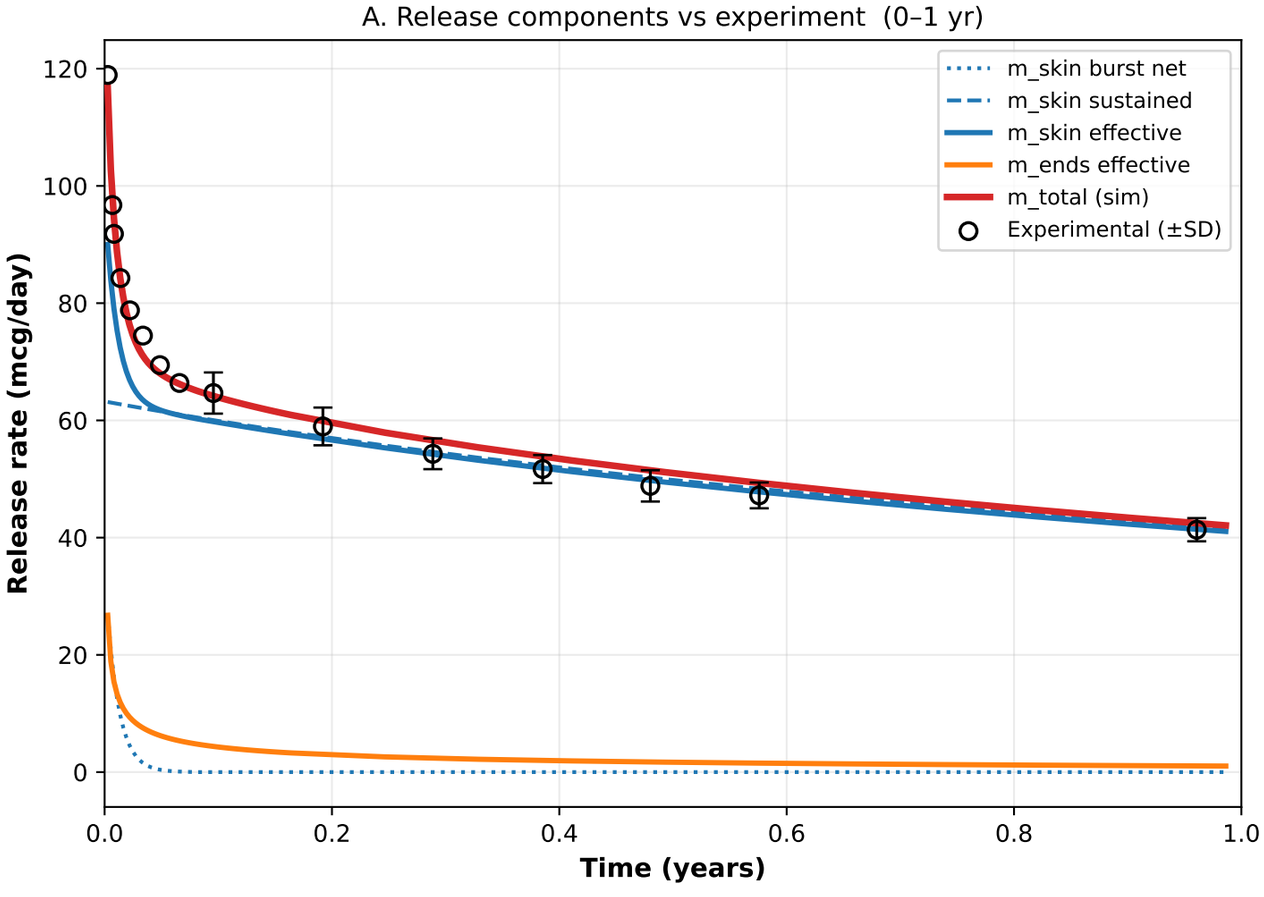
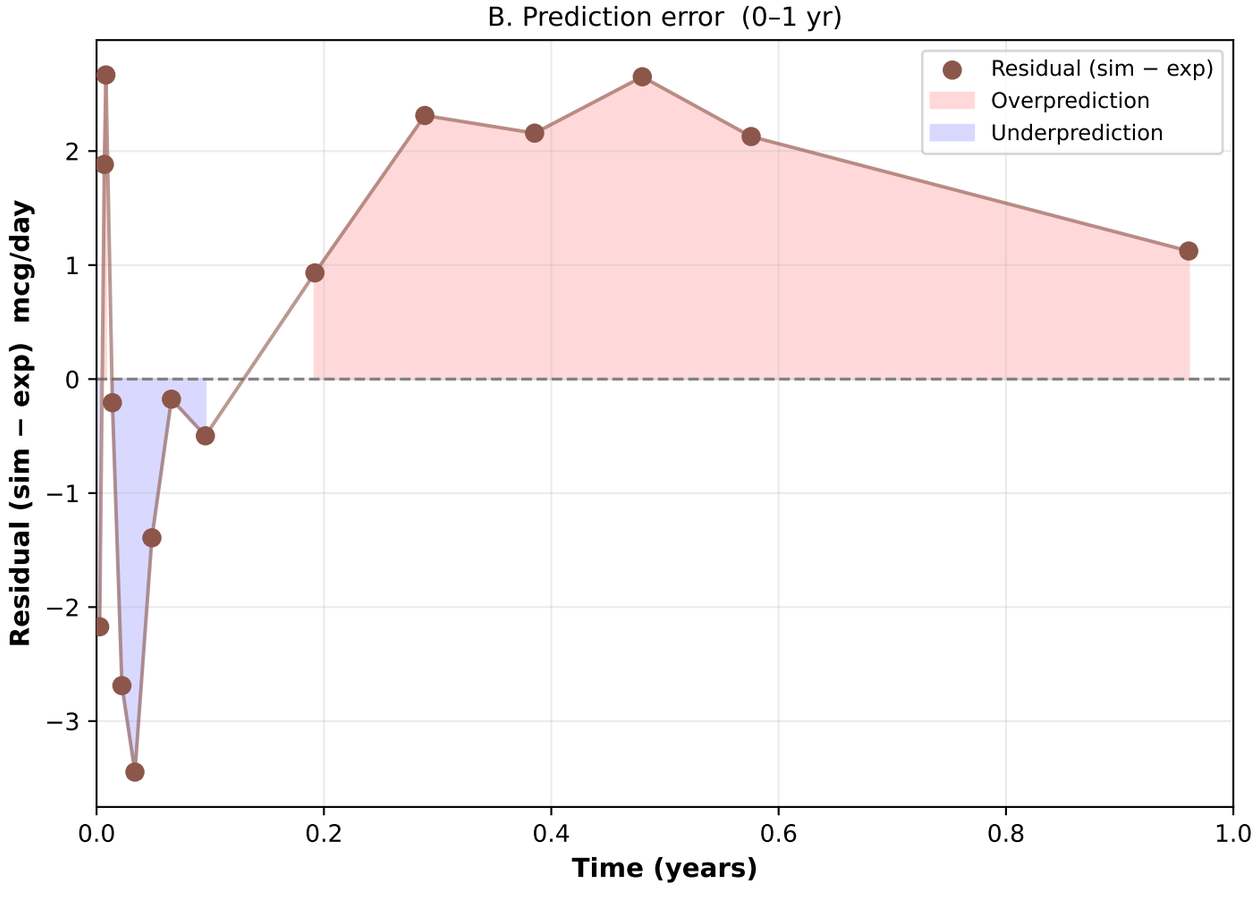


Figure 2. Zoomed view: 0–1 year | Diagnosing early-time discrepancy
DOS=1.535, PDL=9.043e-12 g/(cm·s), ha_skin=0.07808 cm, ha_ends=0.000100 cm





Parameter summary tables

Table 1. Short-term data fitted parameters

Parameter	Value	Unit	Description
DOS	1.535264	-	Degree of supersaturation
PDL	9.042533e-12	g/(cm·s)	Fitted permeability of depletion layer
ha_skin	0.078080	cm	Skin water-layer thickness
ha_ends	0.000100	cm	Ends water-layer thickness

Table 2. Directly measured / fixed parameters (categorized)

Category	Parameter	Value	Unit	Description
Implant geometry	L	4	cm	Implant length
Implant geometry	R_i	0.094	cm	Inner radius
Implant geometry	R_o	0.1	cm	Outer radius
Implant geometry	V_core	0.111036	cm^3	Core volume
Implant geometry	SA_ends	0.0555182	cm^2	Ends surface area
Etonogestrel transport	Cs_28	0.0023	g/cm^3	Solubility in EVA 28
Etonogestrel transport	D_28	3.3e-09	cm^2/s	Diffusivity in EVA 28
Etonogestrel transport	Cs_DL	0.00192	g/cm^3	Solubility in depletion layer
Etonogestrel transport	D_DL	6.36e-09	cm^2/s	Diffusivity in depletion layer
Etonogestrel transport	Cs_15	0.0013	g/cm^3	Solubility in EVA 15
Etonogestrel transport	D_15	2.2e-09	cm^2/s	Diffusivity in EVA 15
Etonogestrel transport	Cs_a	6.9e-06	g/cm^3	Solubility in water layer
Etonogestrel transport	D_a	7.02e-06	cm^2/s	Diffusivity in water layer
Etonogestrel transport	P_DL reference	1.22112e-11	g/(cm·s)	Measured reference used to scale fitted PDL
Etonogestrel transport	P_15	2.86e-12	g/(cm·s)	Permeability of EVA 15
Etonogestrel transport	P_a	4.8438e-11	g/(cm·s)	Permeability of water layer
Initial conditions	Load	68	mg	Initial drug load
Initial conditions	C0	0.612412	g/cm^3	Initial concentration in core
Dissolution medium	sink_condition	0.023	-	C_media / Cs_a
Dissolution medium	C_media	1.587e-07	g/cm^3	Medium concentration
Release baseline	zero_order_ends	0	mcg/day	Constant baseline release from ends

Figure 3. Data-duration sensitivity summary

Table 3. Accuracy on full dataset

Data Duration	# Points	MAE (full)	RMSE (full)	MAPE % (full)	R^2 (full)
3 months	10	5.1362	6.9996	14.4231	0.9215
6 months	13	2.2337	2.9380	5.6345	0.9862
1 year	15	1.9782	2.4761	4.7659	0.9902
2 years	17	1.7291	1.9827	3.6984	0.9937
3 years	19	1.7003	1.9029	3.3705	0.9942

Table 4. Fitted parameters across data durations

Data Duration	DOS	PDL (g/(cm·s))	ha_skin (cm)	ha_ends (cm)
3 months	1.000001	5.301209e-11	0.126953	0.000153
6 months	1.521292	6.072082e-12	0.063144	0.000100
1 year	1.524264	6.778575e-12	0.066703	0.000100
2 years	1.531405	8.132926e-12	0.073604	0.000100
3 years	1.534608	8.896746e-12	0.077359	0.000100

Figure 4. Duration-sensitivity metrics

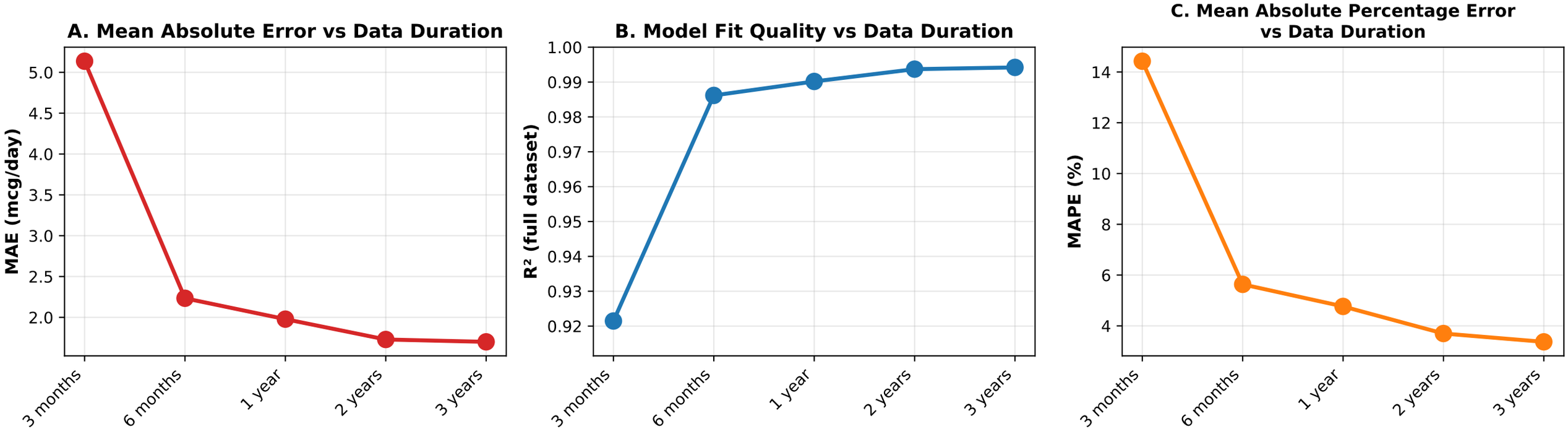


Figure 5. Parameter convergence vs data duration

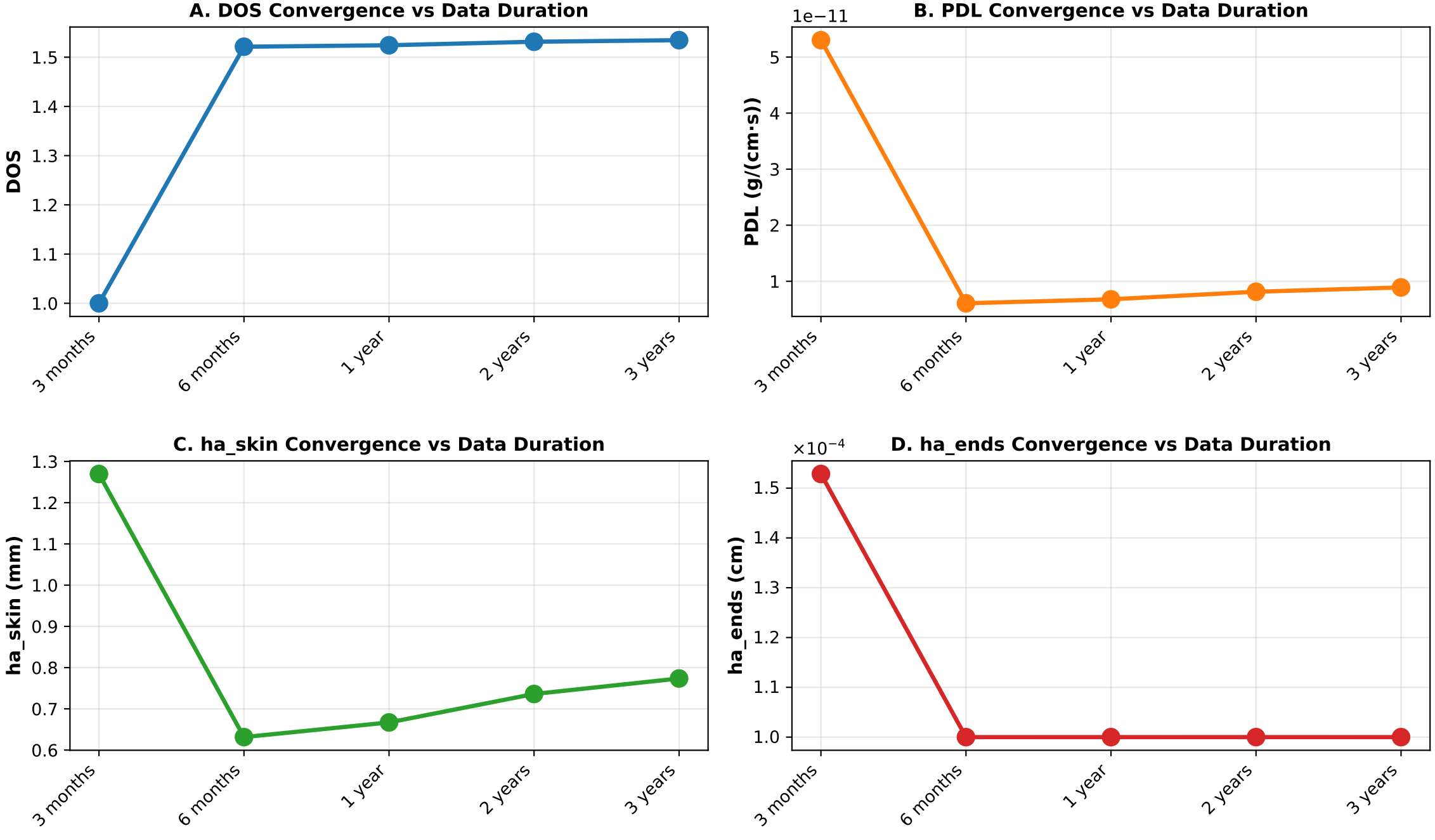


Figure 6. Drug release profiles using PDL fitted from 6, 12, 24, and 36 months data

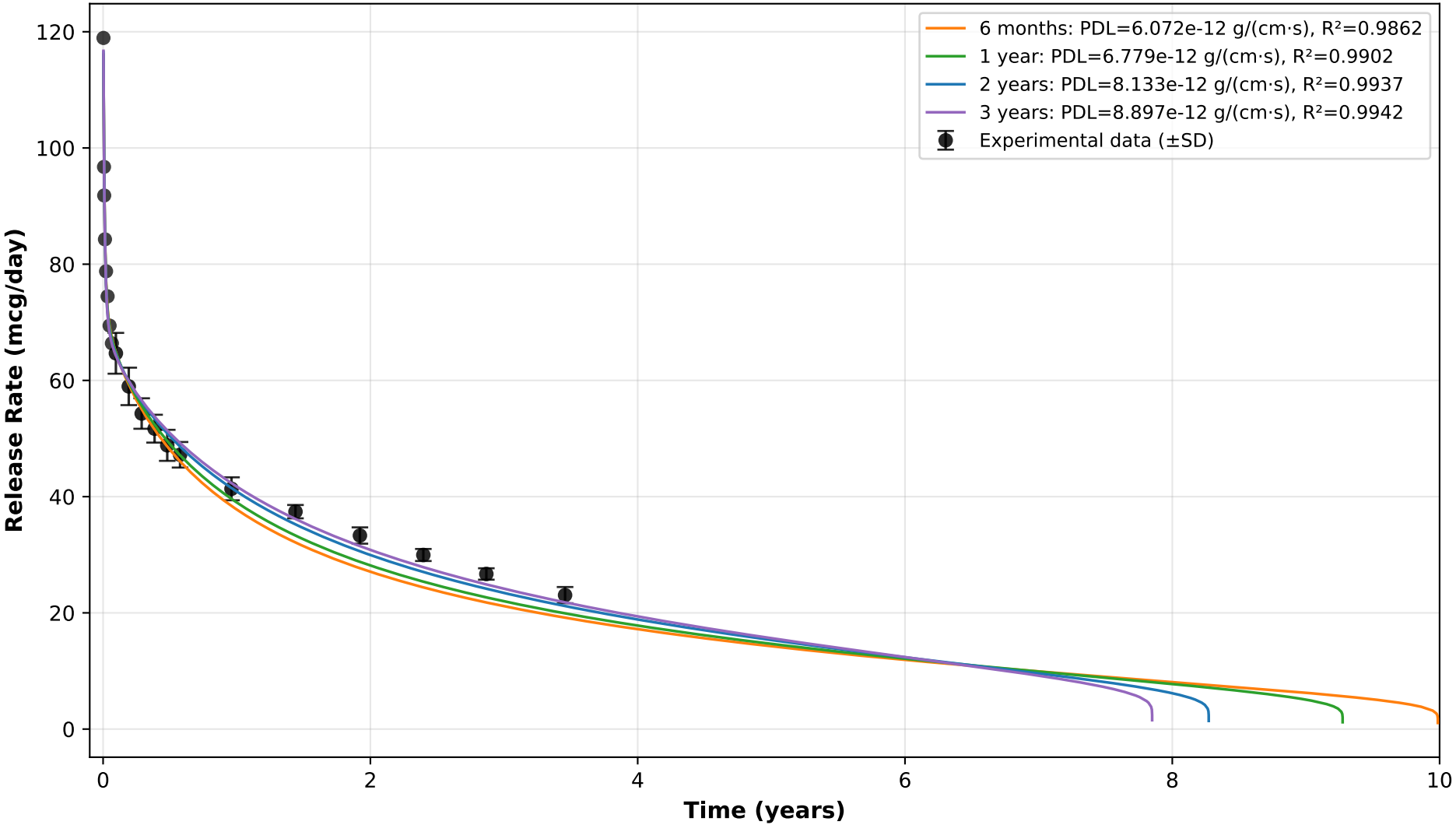


Figure 7. Drug release profiles using parameters fitted from 3 months, 6 months, 1 year, and 3 years data

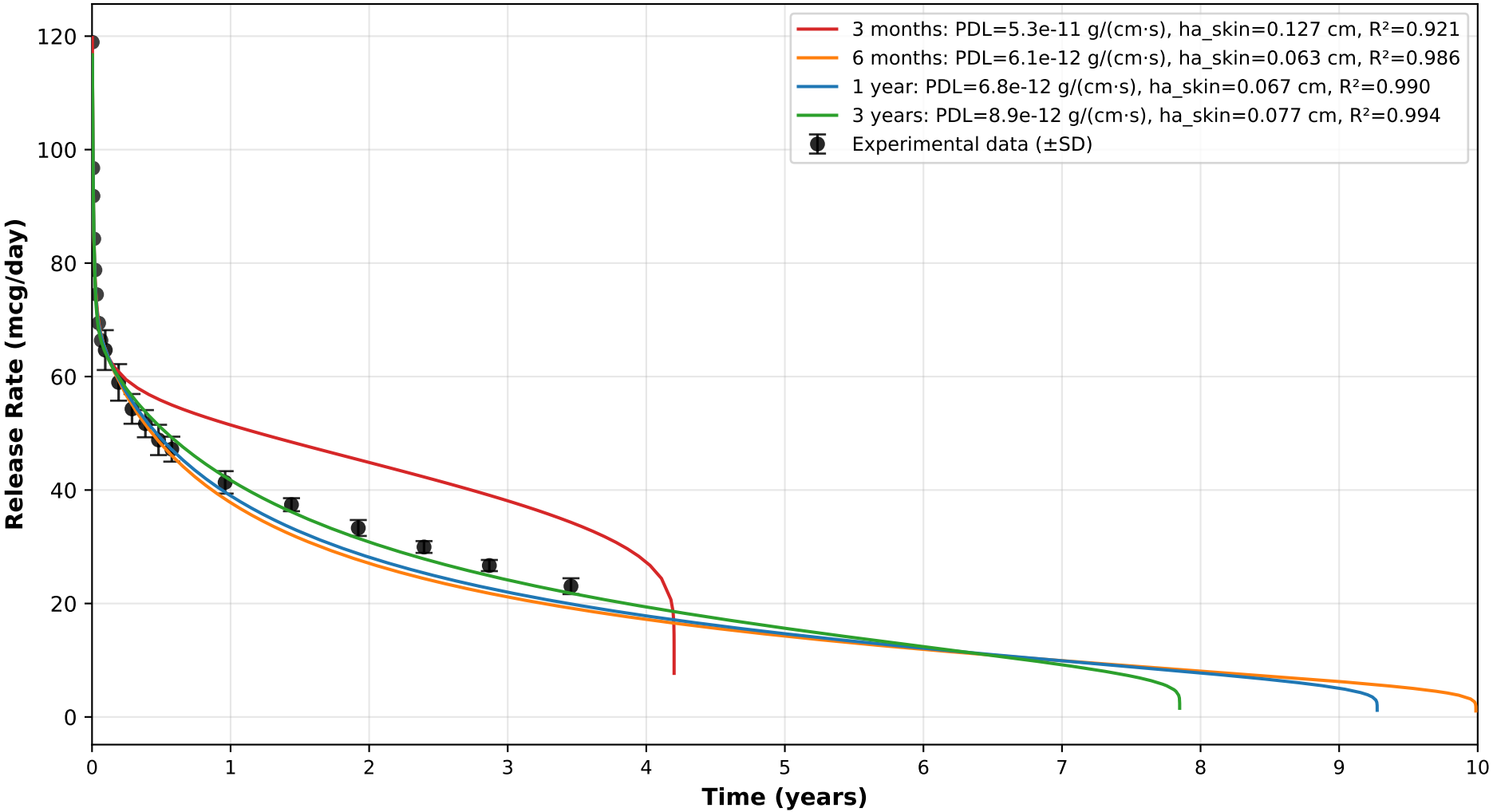
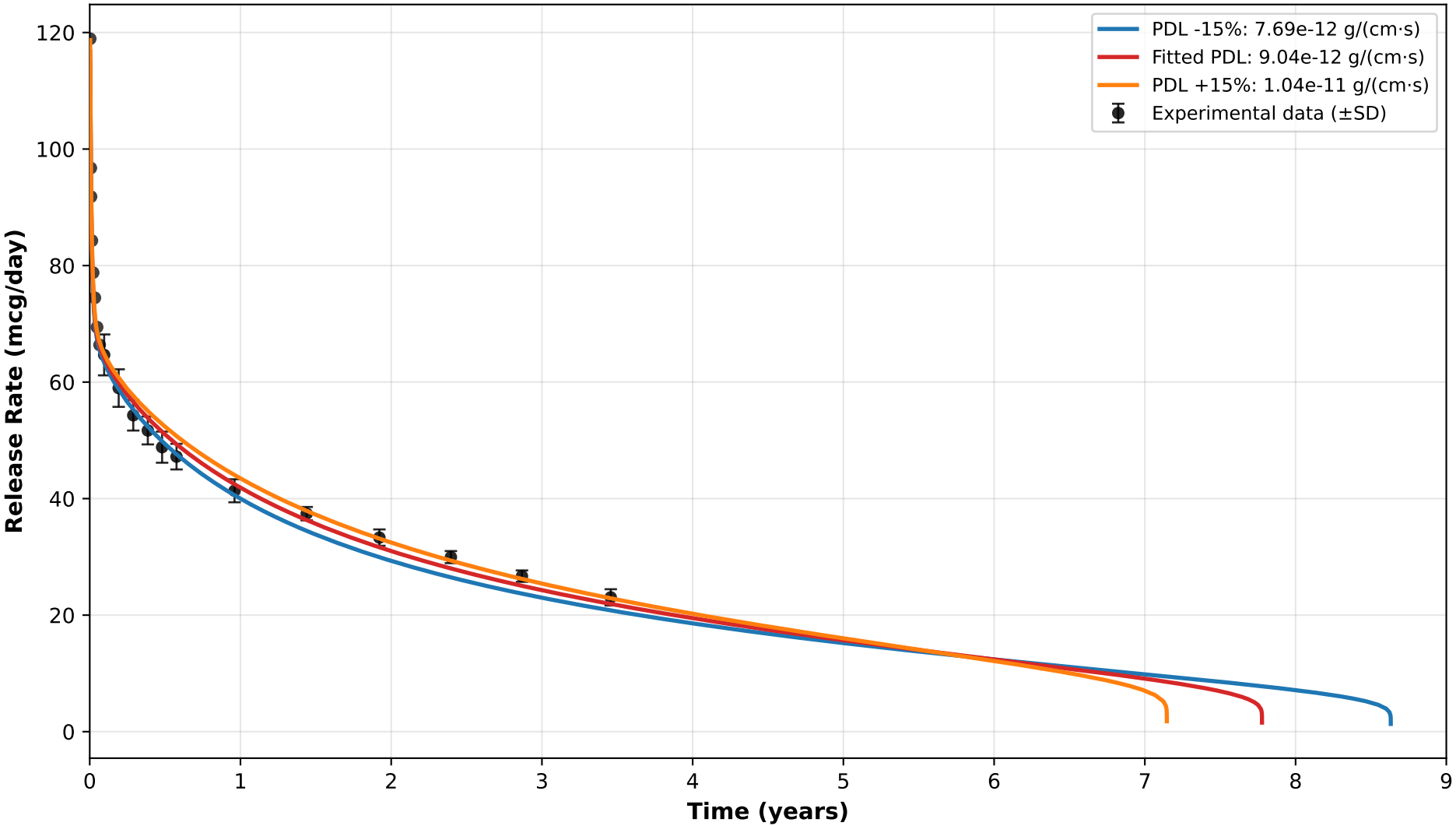


Figure 8. Sensitivity of drug release profiles to fitted PDL ±15%



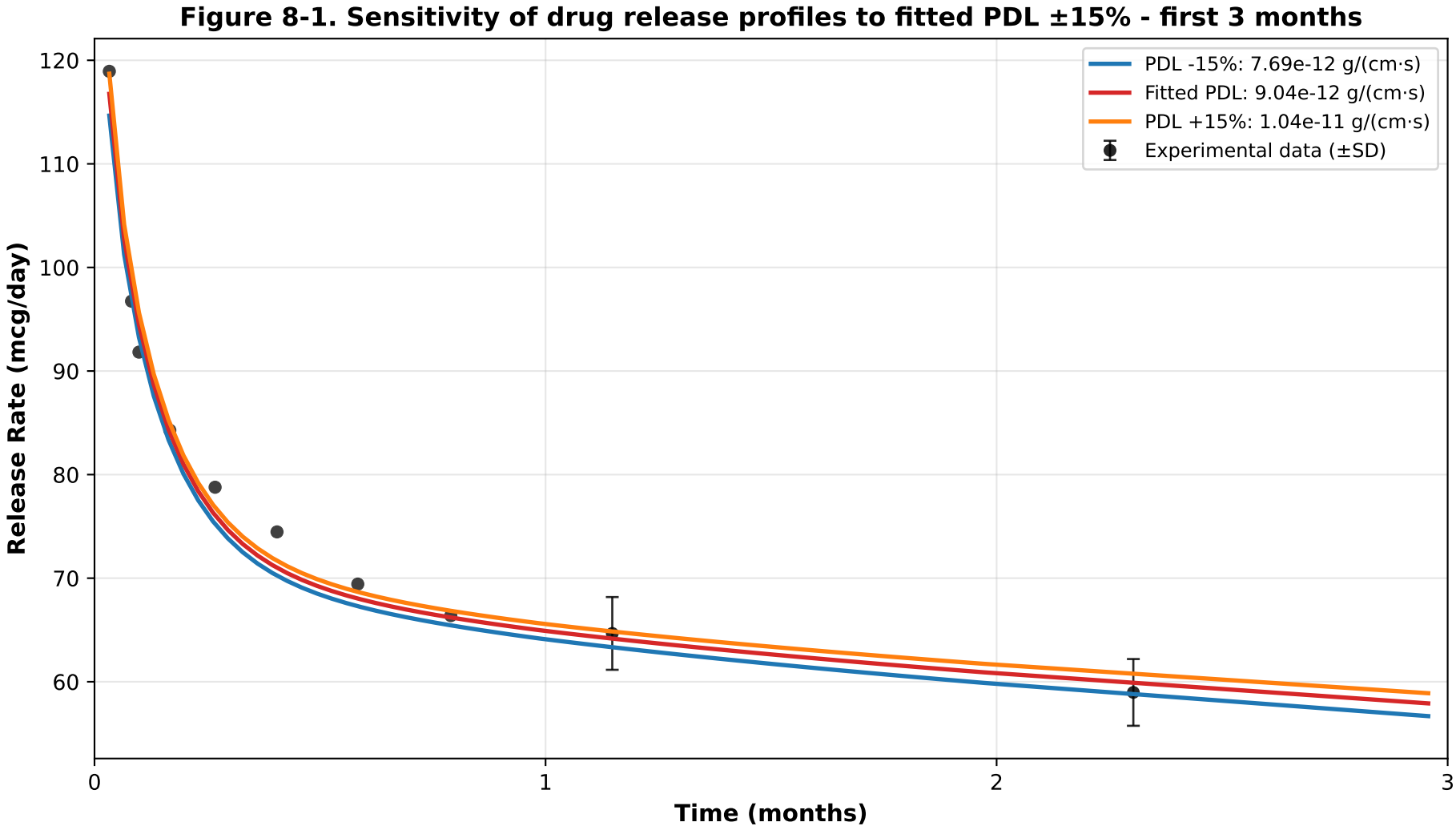

Figure 8-1. Sensitivity of drug release profiles to fitted PDL ±15% - first 3 months
PDL -15%: 7.69e-12 g/(cm·s)
Fitted PDL: 9.04e-12 g/(cm·s)
PDL +15%: 1.04e-11 g/(cm·s)
Experimental data (±SD)
Release Rate (mcg/day)
120
110
100
90
80
70
60
0
1
2
3
Time (months)

Figure 9. Sensitivity of drug release profiles to reference Pskin (P_15) ±15%

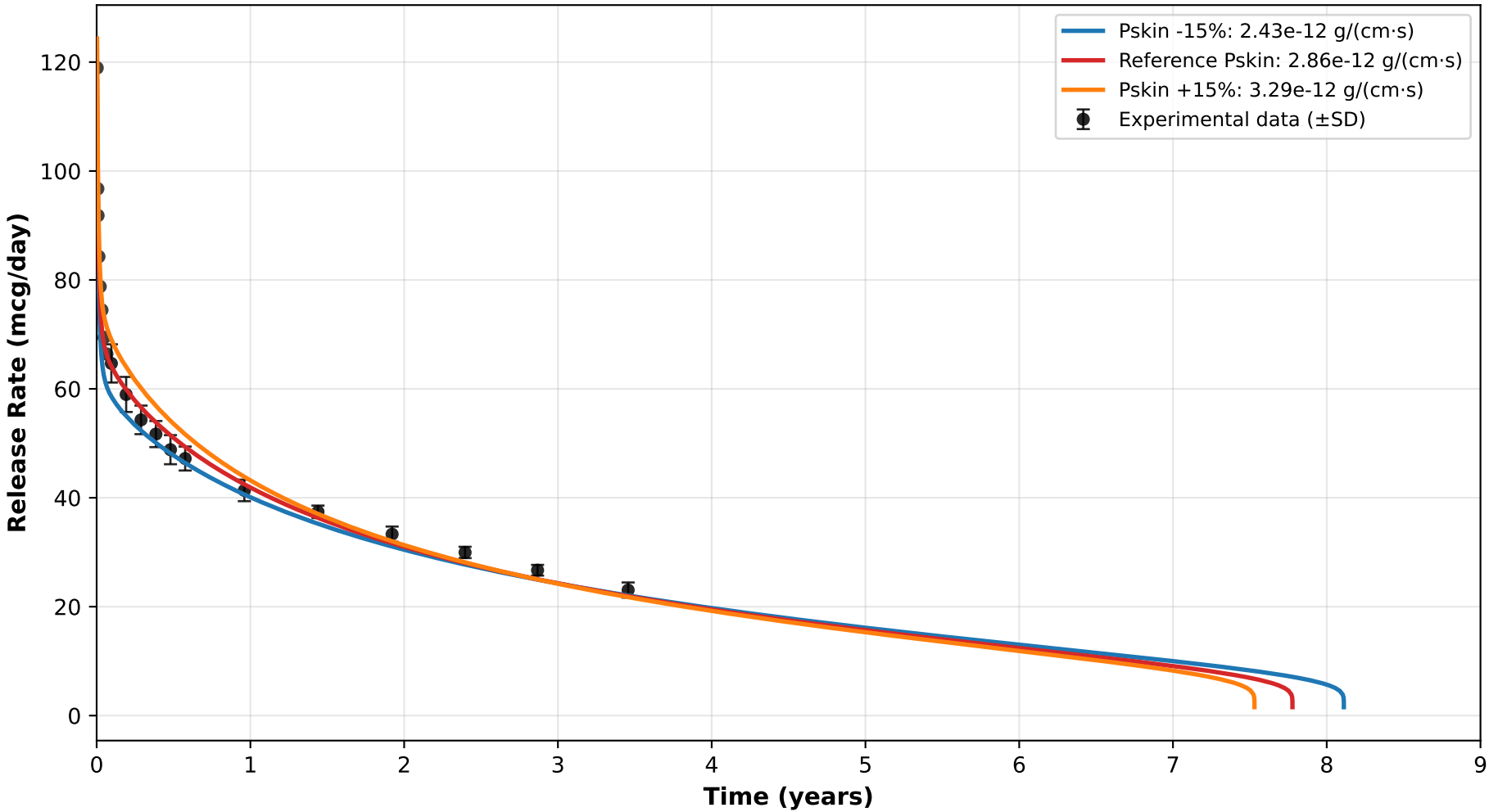
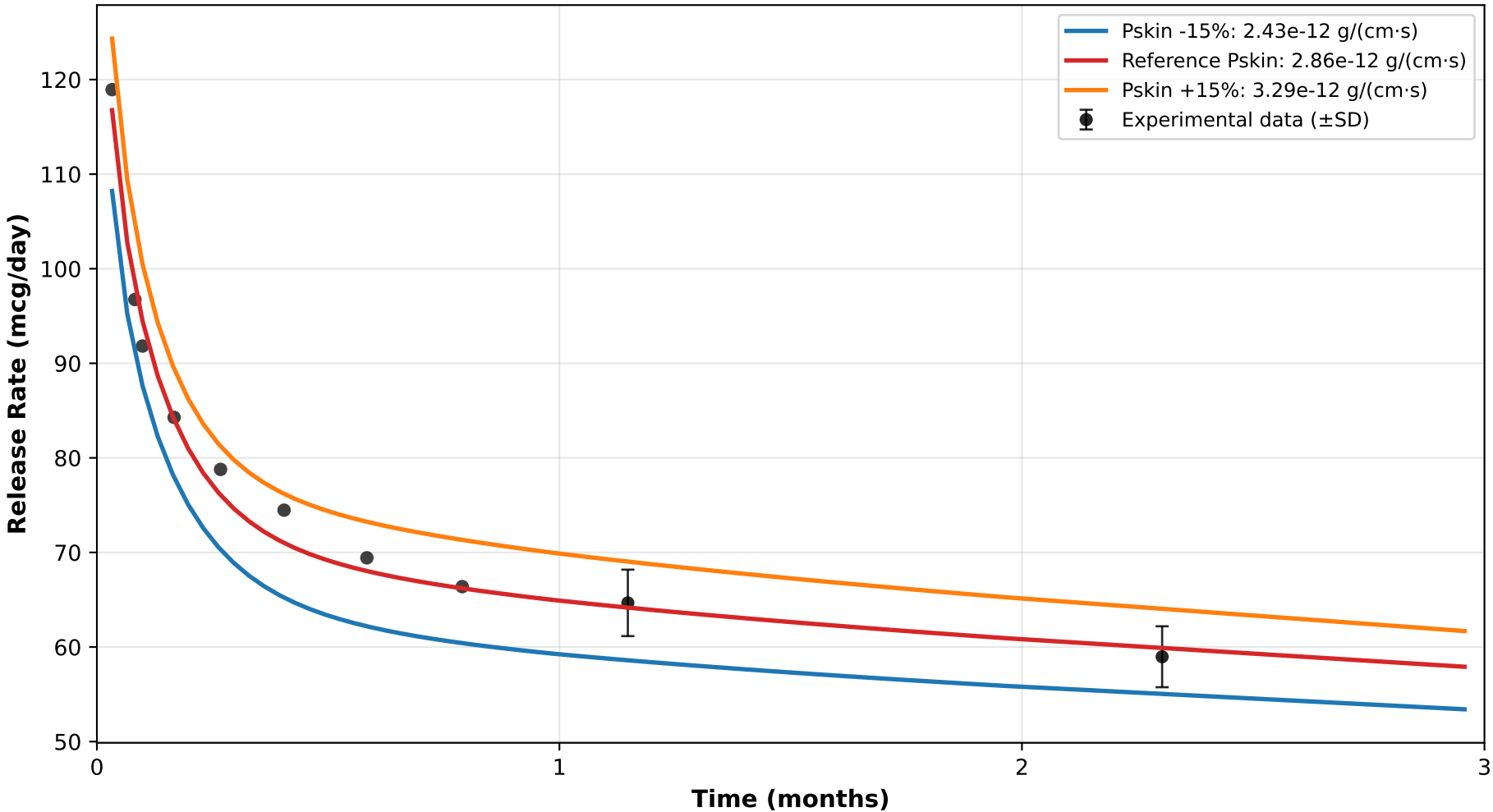
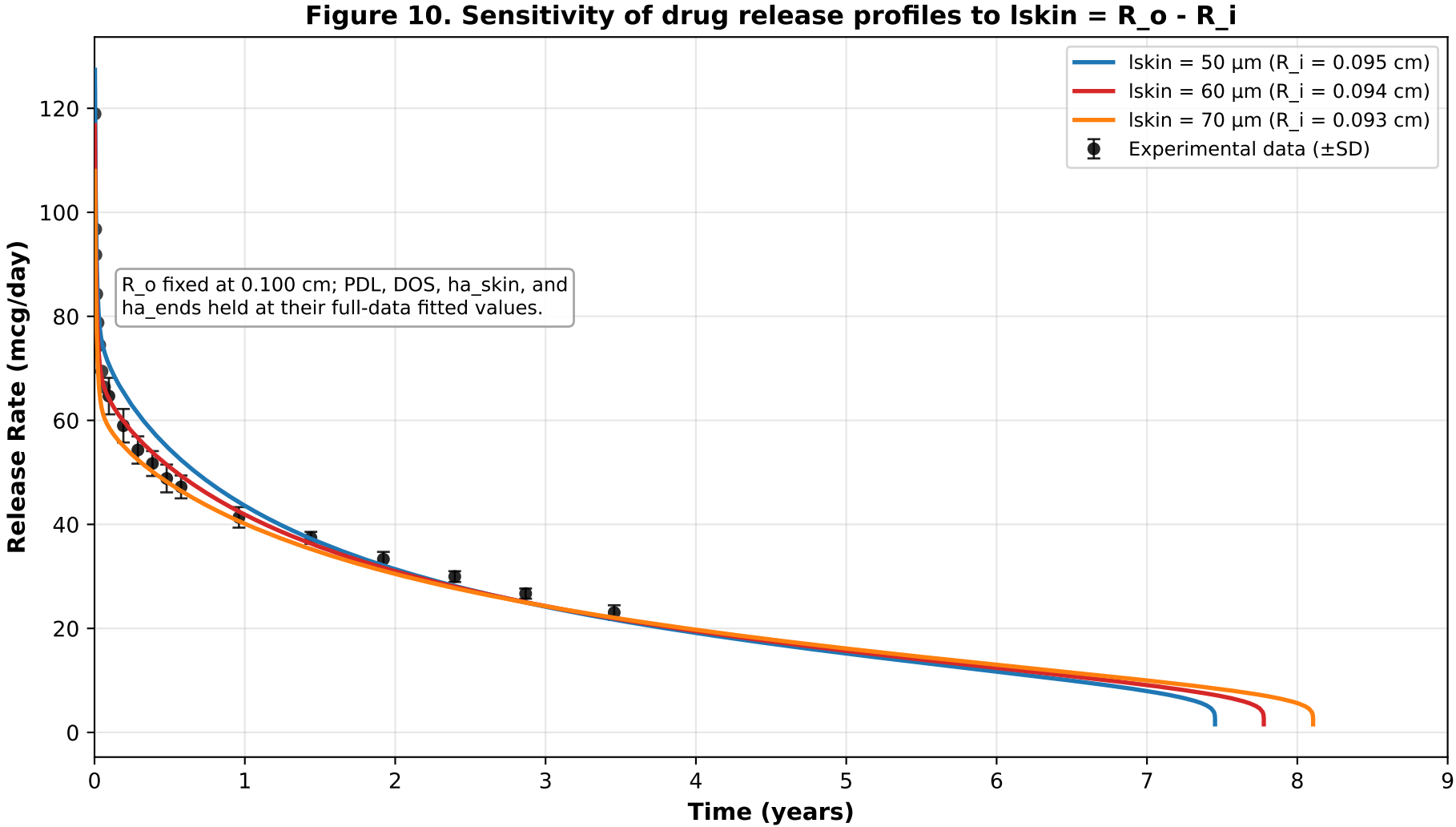


Figure 9-1. Sensitivity of drug release profiles to reference Pskin (P_15) ±15% - first 3 months




Figure 10. Sensitivity of drug release profiles to lskin = R_o - R_i
lskin = 50 µm (R_i = 0.095 cm)
lskin = 60 µm (R_i = 0.094 cm)
lskin = 70 µm (R_i = 0.093 cm)
Experimental data (±SD)
R_o fixed at 0.100 cm; PDL, DOS, ha_skin, and ha_ends held at their full-data fitted values.
Release Rate (mcg/day)
Time (years)
0
20
40
60
80
100
120
0
1
2
3
4
5
6
7
8
9

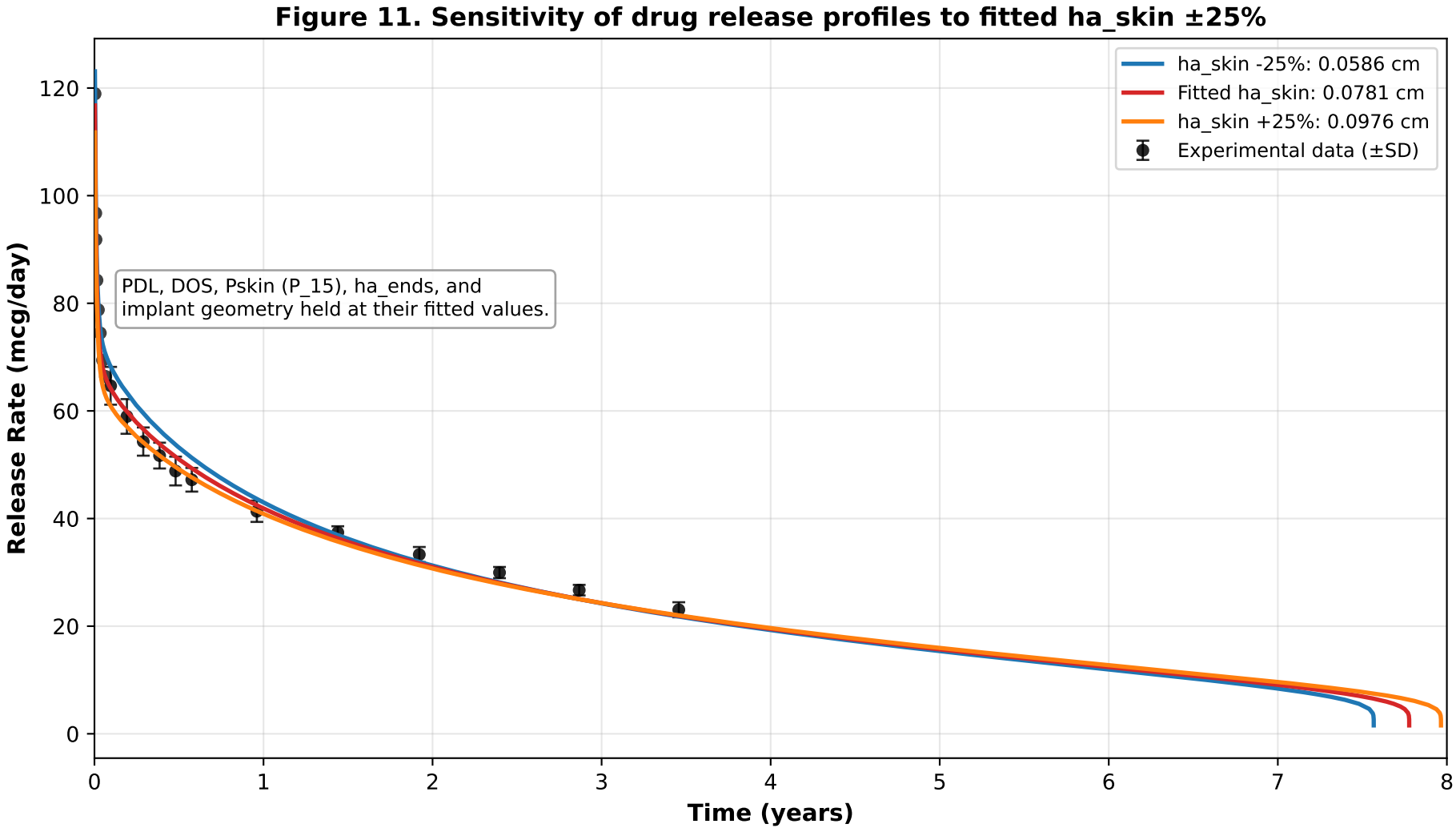
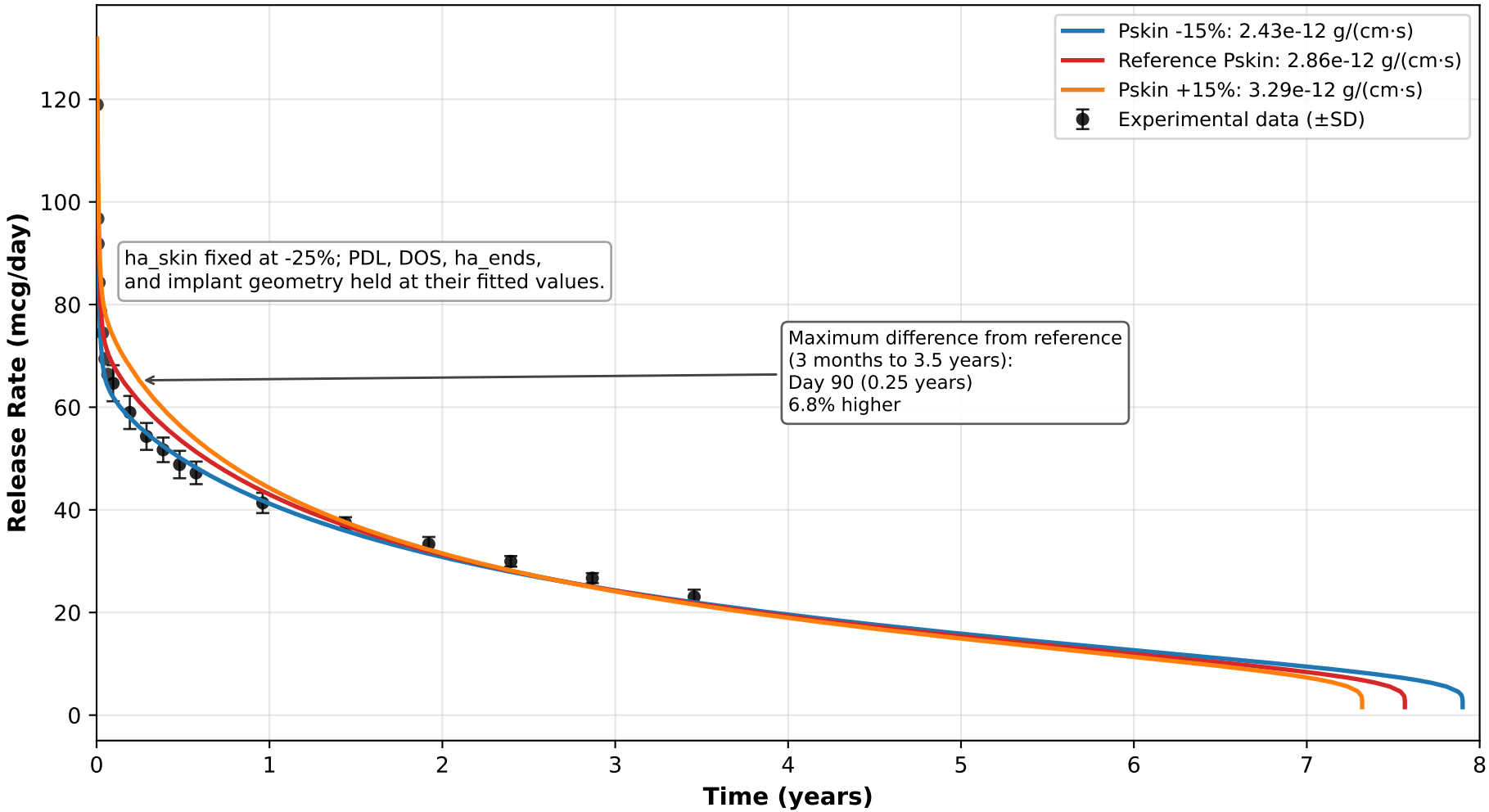
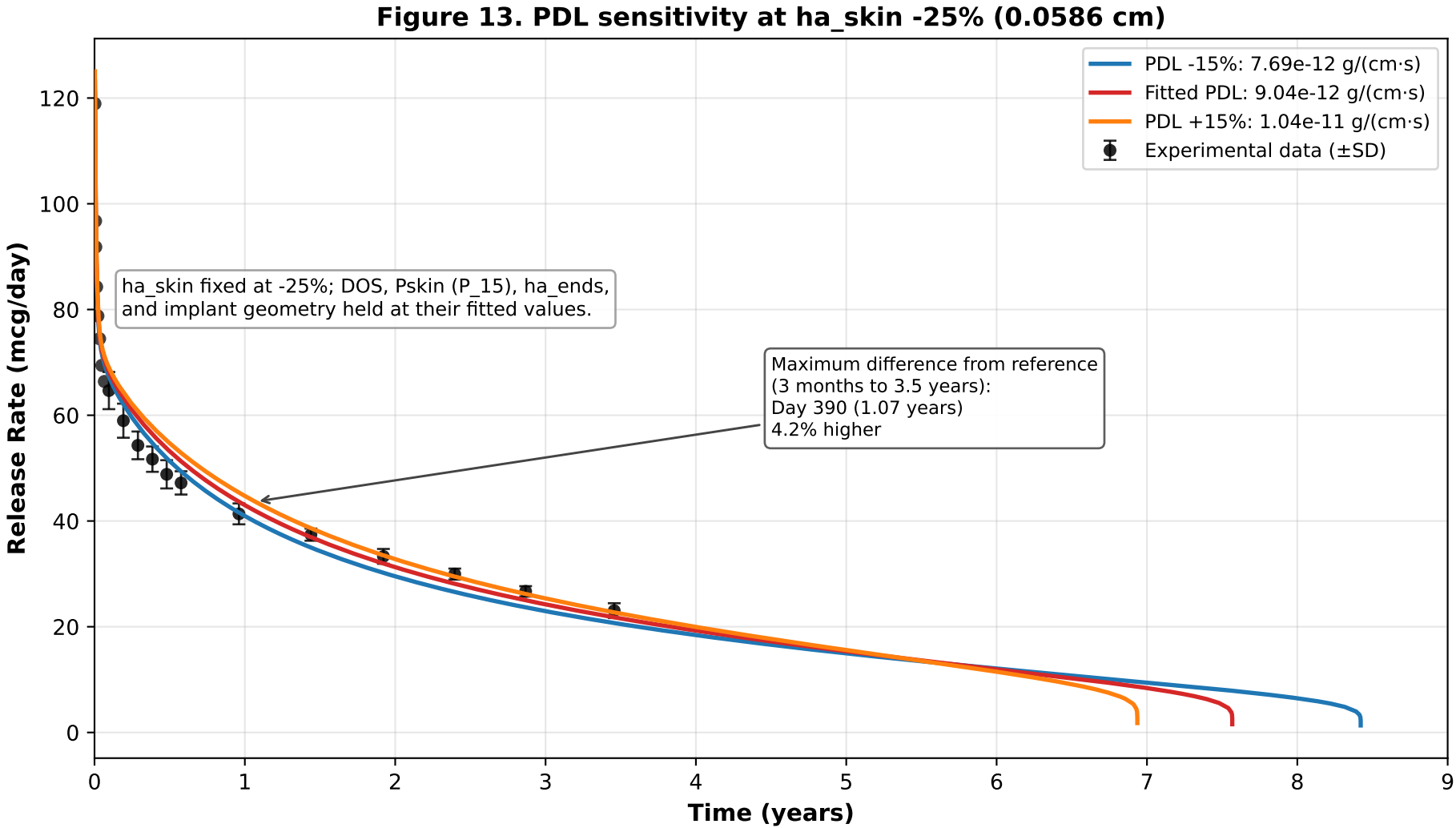
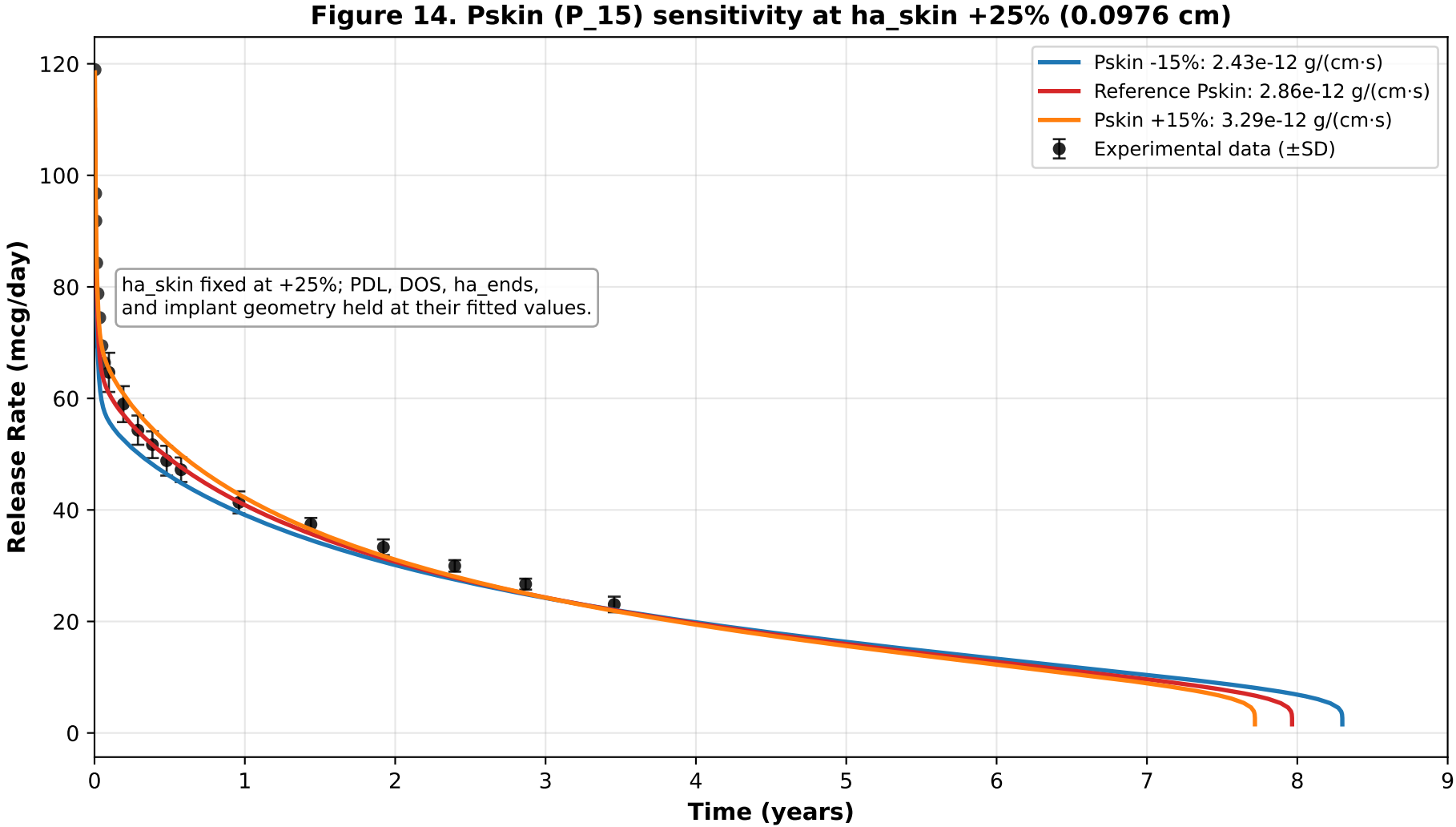

Figure 11. Sensitivity of drug release profiles to fitted ha_skin ±25%
ha_skin -25%: 0.0586 cm
Fitted ha_skin: 0.0781 cm
ha_skin +25%: 0.0976 cm
Experimental data (±SD)
PDL, DOS, Pskin (P_15), ha_ends, and implant geometry held at their fitted values.
Release Rate (mcg/day)
Time (years)
0
20
40
60
80
100
120
0
1
2
3
4
5
6
7
8

Figure 12. Pskin (P_15) sensitivity at ha_skin -25% (0.0586 cm)




Figure 13. PDL sensitivity at ha_skin -25% (0.0586 cm)
PDL -15%: 7.69e-12 g/(cm·s)
Fitted PDL: 9.04e-12 g/(cm·s)
PDL +15%: 1.04e-11 g/(cm·s)
Experimental data (±SD)
ha_skin fixed at -25%; DOS, Pskin (P_15), ha_ends,
and implant geometry held at their fitted values.
Maximum difference from reference
(3 months to 3.5 years):
Day 390 (1.07 years)
4.2% higher
Release Rate (mcg/day)
Time (years)
0
20
40
60
80
100
120
0
1
2
3
4
5
6
7
8
9


Figure 14. Pskin (P_15) sensitivity at ha_skin +25% (0.0976 cm)
Pskin -15%: 2.43e-12 g/(cm·s)
Reference Pskin: 2.86e-12 g/(cm·s)
Pskin +15%: 3.29e-12 g/(cm·s)
Experimental data (±SD)
ha_skin fixed at +25%; PDL, DOS, ha_ends,
and implant geometry held at their fitted values.
Release Rate (mcg/day)
Time (years)
0
20
40
60
80
100
120
0
1
2
3
4
5
6
7
8
9

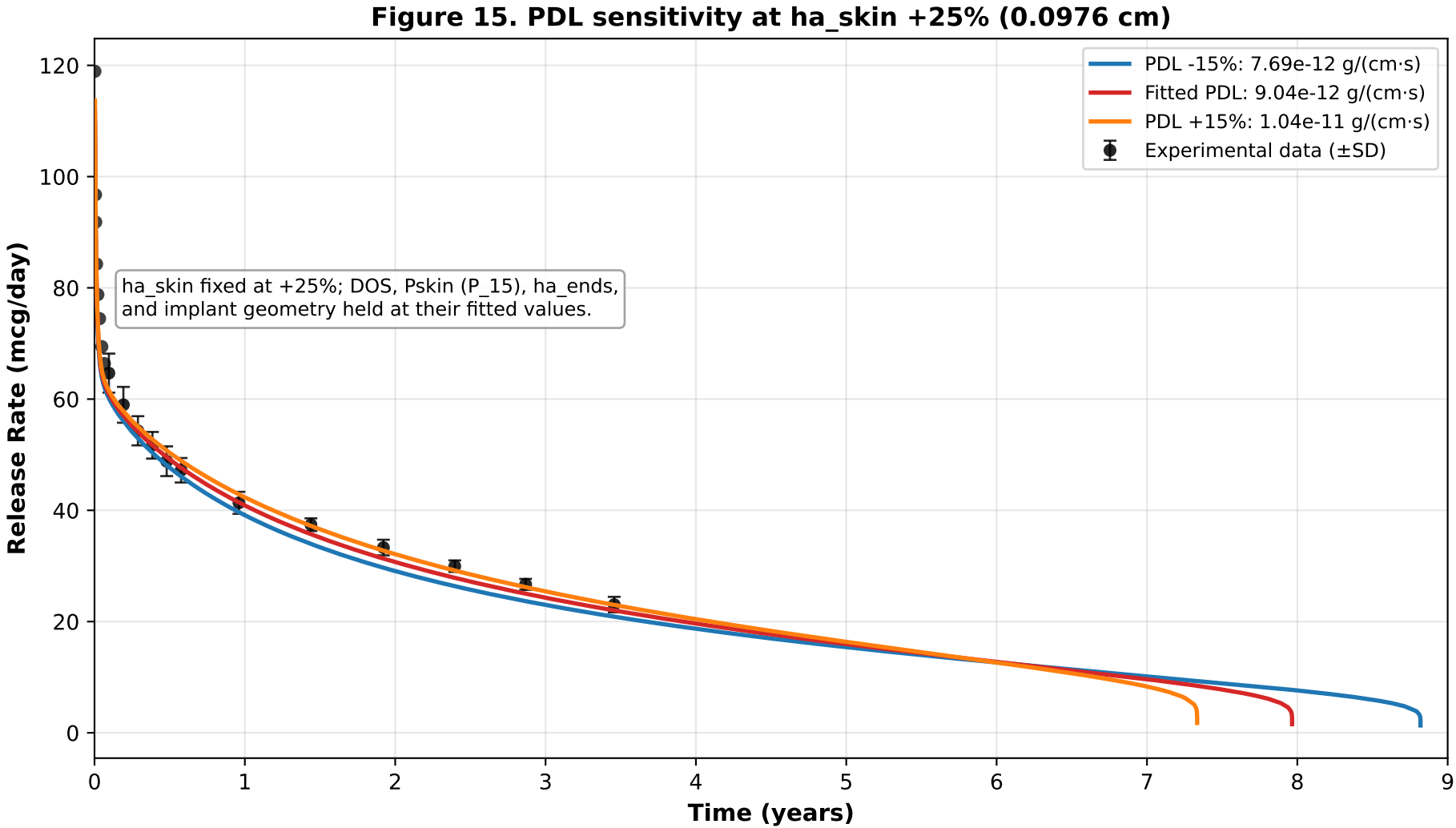

Figure 15. PDL sensitivity at ha_skin +25% (0.0976 cm)
PDL -15%: 7.69e-12 g/(cm·s)
Fitted PDL: 9.04e-12 g/(cm·s)
PDL +15%: 1.04e-11 g/(cm·s)
Experimental data (±SD)
ha_skin fixed at +25%; DOS, Pskin (P_15), ha_ends,
and implant geometry held at their fitted values.
Release Rate (mcg/day)
Time (years)
0
20
40
60
80
100
120
0
1
2
3
4
5
6
7
8
9

Figure 16. Impact of crossed PDL/Pskin changes on drug release profiles

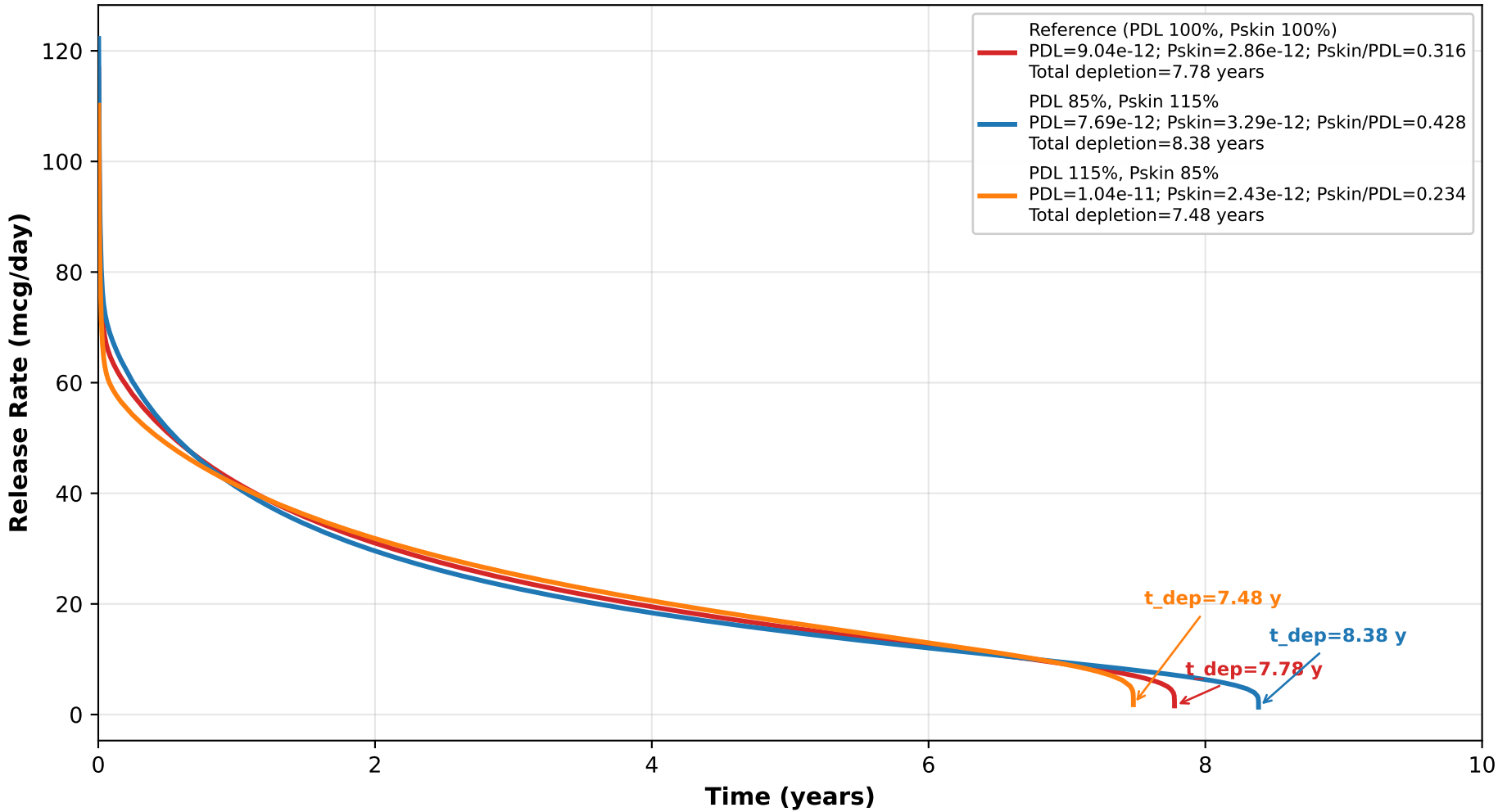
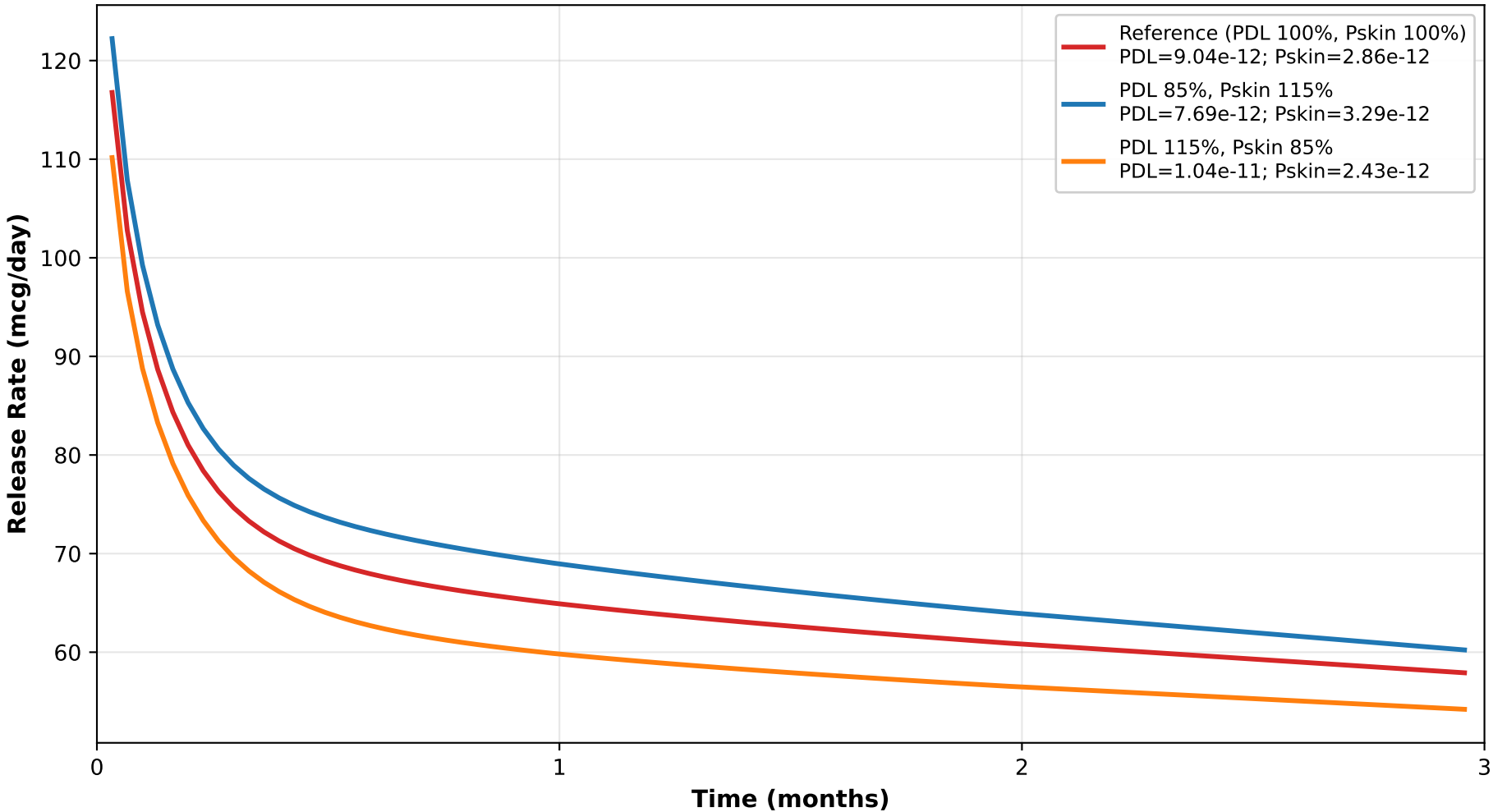


Figure 16-1. Crossed PDL/Pskin changes - first 3 months



Appendix A. Release-model equations (1 of 2)

Rates are calculated in g/s and then converted to mcg/day (×10^6 × 86,400). The equations below use the fitted values of DOS, PDL, ha_skin, and ha_ends.

1. Burst release through the skin (first 30 days)

$$R_{a,skin} = R_o + h_{a,skin}, \quad C_{core,0} = DOS\, C_{s,28}, \quad P_{15} = C_{s,15} D_{15}, \quad P_a = C_{s,a} D_a$$

$$C_{1,burst} = \frac{2\pi L}{\ln(R_o/R_i)/P_{15} + \ln(R_{a,skin}/R_o)/P_a}$$

$$C_{2,burst} = 1 - C_{media}/C_{s,a}, \quad C_{3,burst} = C_{core,0}/C_{s,28} - 1$$

$$m_{burst}(t) = C_{1,burst}[C_{2,burst} + C_{3,burst} \exp\{-C_{1,burst} t/(C_{s,28} V_{core})\}]$$

2. Radial depletion position used for sustained skin release

$$C_{1,R_d} = -(1-s)/(C_0 - C_{s,DL}), \quad C_{2,R_d} = 1/PDL$$

$$C_{4,R_d} = \ln(R_o/R_i)/P_{15} + \ln(R_{a,skin}/R_o)/P_a$$

$$A = C_{2,R_d}(2\ln R_i + 1) + 2C_{4,R_d}$$

$$t(R_d) = \frac{(R_d^2 - R_i^2)A + 2C_{2,R_d}[-R_d^2 \ln R_d + R_i^2 \ln R_i]}{4C_{1,R_d}}$$

3. Sustained skin release

$$m_{skin,sus}(t) = \frac{2\pi L(1-s)}{\ln(R_i/R_d(t))/PDL + \ln(R_o/R_i)/P_{15} + \ln(R_{a,skin}/R_o)/P_a}$$

At each simulated time, R_d is obtained by numerically solving the displayed t(R_d) relation. The model includes radius-spaced points from 5% R_i to 0.001% R_i to resolve late depletion.

Appendix A. Release-model equations and variables (2 of 2)

4. End release

$C_{1,end} = -(1-s)/(C_0 - C_{s,DL}), \quad C_{2,end} = L/(2PDL) + h_{a,end}/P_a, \quad C_{3,end} = -1/PDL$

$$t = \frac{0.5C_{3,end}Z_d^2 + C_{2,end}Z_d - C_{2,end}L/2 - C_{3,end}L^2/8}{C_{1,end}}$$

$$m_{end}(t) = \frac{(1-s)SA_{ends}}{[L/2 - Z_d(t)]/PDL + h_{a,end}/P_a} + m_{end,0}$$

5. Combining burst, skin, and ends release

$m_{burst,net} = \max\{m_{burst} - m_{skin,sus} - (t/30)m_{burst,extra}(30), 0\}$ for $t \le 30$ days; otherwise 0

$m_{skin} = m_{skin,sus} + m_{burst,net}, \quad Q_j(t) = \int_0^t m_j(\tau)\,d\tau$ (numerical trapezoidal integration)

$m_{skin,eff} = m_{skin}\max(1 - Q_{end}/Load, 0), \quad m_{end,eff} = m_{end}\max(1 - Q_{skin}/Load, 0)$

$Q_{total}(t) = \int_0^t m_{total}(\tau)\,d\tau$

Variables and fixed inputs

Geometry: L = implant length; R_i/R_o = inner/outer radius; R_a,skin = outer aqueous boundary; R_d = radial depletion front; Z_d = end depletion front; V_core = πR_i²L; SA_ends = 2πR_i²; h_a,skin and h_a,end = aqueous boundary-layer thicknesses.

Transport/concentration: C_s,28, C_s,DL, C_s,15, C_s,a = solubilities; D_28, D_DL, D_15, D_a = diffusivities; PDL = fitted drug-layer permeability; P_15 = Pskin = C_s,15D_15; P_a = C_s,aD_a; C_0 = initial core concentration; C_media = sC_s,a; s = sink condition.

Release/output: DOS = fitted initial concentration multiplier; t = time (s in equations, days in output); m = instantaneous release rate; Q = cumulative released amount; Load = 68 mg initial drug load; m_end,0 = zero-order end term (0 mcg/day here); subscripts burst, skin, end, sus, eff, and total indicate the corresponding component.

The effective rates prevent either pathway from releasing more drug than remains after the other pathway's cumulative release. Thus the table reports Q_skin,eff + Q_end,eff = Q_total at every time point (within numerical precision).

Appendix B. Quarterly simulated release data (1 of 2)

Rates are instantaneous fitted-model rates. Drug amounts are cumulative effective released mass; the final row is the simulated depletion endpoint.

Time (months)	Skin release rate (mcg/day)	Effective skin release rat (mcg/day)	Skin release drug amoun (mg)	Ends release rate (mcg/day)	Effective ends release ra (mcg/day)	Ends release drug amour (mg)	Total release rate (mcg/day)	Total release drug amoun (mg)
3	55.619	55.221	5.517	2.808	2.580	0.472	57.801	5.990
6	49.833	49.320	10.279	1.984	1.683	0.661	51.003	10.940
9	45.237	44.663	14.562	1.620	1.270	0.794	45.934	15.356
12	41.466	40.856	18.461	1.403	1.019	0.898	41.875	19.359
15	38.294	37.662	22.042	1.255	0.844	0.982	38.506	23.024
18	35.572	34.928	25.353	1.145	0.713	1.053	35.642	26.406
21	33.199	32.549	28.432	1.060	0.612	1.114	33.161	29.545
24	31.103	30.451	31.306	0.992	0.529	1.166	30.980	32.472
27	29.229	28.579	34.000	0.935	0.461	1.211	29.040	35.210
30	27.539	26.892	36.531	0.887	0.404	1.250	27.296	37.781
33	26.000	25.360	38.916	0.846	0.354	1.285	25.714	40.200
36	24.588	23.956	41.166	0.810	0.312	1.315	24.267	42.482
39	23.285	22.661	43.294	0.778	0.275	1.342	22.935	44.636
42	22.073	21.458	45.308	0.750	0.242	1.365	21.700	46.673
45	20.939	20.336	47.216	0.724	0.213	1.386	20.548	48.602
48	19.873	19.281	49.024	0.701	0.187	1.404	19.468	50.428

Skin release rate includes sustained release plus the tapered 0–30-day burst contribution. Effective rates apply the cross-pathway depletion factor used by the model.

Appendix B. Quarterly simulated release data (2 of 2)

Rates are instantaneous fitted-model rates. Drug amounts are cumulative effective released mass; the final row is the simulated depletion endpoint.

Time (months)	Skin release rate (mcg/day)	Effective skin release rat (mcg/day)	Skin release drug amoun (mg)	Ends release rate (mcg/day)	Effective ends release ra (mcg/day)	Ends release drug amour (mg)	Total release rate (mcg/day)	Total release drug amoun (mg)
51	18.865	18.286	50.739	0.680	0.163	1.420	18.449	52.159
54	17.906	17.340	52.365	0.661	0.143	1.434	17.483	53.800
57	16.989	16.438	53.907	0.643	0.124	1.446	16.561	55.354
60	16.108	15.571	55.369	0.627	0.107	1.457	15.678	56.825
63	15.255	14.734	56.752	0.612	0.091	1.466	14.825	58.218
66	14.425	13.920	58.060	0.598	0.077	1.474	13.997	59.534
69	13.611	13.124	59.295	0.585	0.064	1.480	13.188	60.775
72	12.805	12.337	60.457	0.572	0.053	1.485	12.390	61.943
75	11.999	11.551	61.548	0.561	0.043	1.490	11.594	63.038
78	11.182	10.756	62.566	0.550	0.033	1.493	10.789	64.060
81	10.338	9.937	63.511	0.540	0.025	1.496	9.961	65.007
84	9.440	9.067	64.379	0.530	0.017	1.498	9.084	65.877
87	8.437	8.097	65.164	0.521	0.011	1.499	8.108	66.663
90	7.189	6.895	65.851	0.512	0.005	1.500	6.900	67.350
93	4.616	4.424	66.396	0.504	0.001	1.500	4.425	67.896
93.26	1.624	1.556	66.428	0.503	0.001	1.500	1.556	67.928

Skin release rate includes sustained release plus the tapered 0–30-day burst contribution. Effective rates apply the cross-pathway depletion factor used by the model.

Appendix C. Estimated medium concentration: 0-5 years

The calculation uses the fitted total-release profile in Figure 1E through year 5. The fit itself uses 3.5 years of experimental data.

1. Area under the fitted release-rate curve

$$AUC_{0-5y} = \int_0^{1825\ \text{days}} m_{total}(t)\,dt$$

= 57,931.9 mcg = 57.932 mg

2. Average release rate over 5 years

$$\overline{m} = AUC_{0-5y}/1825\ \text{days}$$

= 31.744 mcg/day

3. End-of-day concentration for 100 mL medium replaced daily

$$M_{daily} = \overline{m} \times 1\ \text{day}$$

= 31.744 mcg per daily medium sample

$$C_{end} = \overline{M}_{daily}/100\ \text{mL}$$

= 0.3174 mcg/mL

4. Time-averaged daily medium concentration

$$\overline{C}_{medium} = \frac{1}{1\ \text{day}}\int_0^{1\ \text{day}} \frac{\overline{m}\,t}{100\ \text{mL}}dt = C_{end}/2$$

= 0.1587 mcg/mL

Relation to the sink-condition definition

Water solubility, C_s,a = 6.900 mcg/mL. Average sink ratio = C_medium/C_s,a = 0.0230 (2.30%).

The model uses a sink condition of 0.023 (2.3%). The corrected value is the time average over each daily replacement interval, assuming constant release.